

Harshan A M

Full-Stack AI Engineer

ON-PREM & AGENTIC AI · IOT · VOICE

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◆ CORE FOCUS

- Agentic AI
- RAG
- Voice AI
- Industrial IoT
- Distributed Systems
- Full-Stack

00 Summary

Full-Stack AI Engineer with **5.5 years** shipping AI-powered, real-time, and scalable systems across EdTech, Industrial IoT, AgriTech, and enterprise tooling for aerospace manufacturing. Depth in LLM systems and **RAG (naive → agentic)**, on-prem / air-gapped **agentic AI** (LangGraph, PydanticAI, MCP, local LLMs), NLP, Voice AI (ASR/TTS), and data-intensive architectures (Kafka, TimescaleDB, time-series). End-to-end ownership — UX/UI and system design through production — in JavaScript, TypeScript, and Python.

01 Experience

Lingotran · Rangsons Group

JAN 2023 – PRESENT · MYSORE
Software Engineer — Full Stack & AI Systems

Core platform — EdTech AI (Lingotran)

- Built LLM-powered AI tutor & recommendation systems (OpenAI, Gemini, Ollama) across multiple language tracks
- Designed production **RAG** pipelines (LangChain, LangGraph, Chroma) with HyDE, RAPTOR, reranking (BGE-M3, Cohere)
- Real-time conversational AI (LiveKit + WebSockets); ASR/TTS via Whisper, Deepgram, Kokoro; pronunciation scoring
- Automated AI-generated learner performance reports; gamified modules with real-time feedback; dynamic CMS audio via ElevenLabs
- LangFuse observability, MCP server integrations, SpaCy NLP (PT/ES/FR); Scrum leader & intern mentoring

◆ SELECTED PLATFORMS DELIVERED ACROSS THE GROUP · PROPRIETARY, NO PUBLIC REPOSITORIES

Graylinx — OMNYX & THERMYNX INDUSTRIAL IOT · AGENTIC AI

- Architected **OMNYX**, an on-prem air-gapped IoT platform: BACnet → Kafka → dual-DB (PostgreSQL 16 + TimescaleDB) with **zero cloud data egress**
- Owned its agentic-AI layer — Planner/Executor/Validator on a local LLM (Ollama/vLLM) with pgvector RAG and Keycloak-modelled agent authorization
- Built **THERMYNX**, the shipping HVAC vertical (FastAPI + React): 5 autonomous tool-calling agents with SSE-streamed reasoning traces
- Architected Digital-Twin FDD/RUL and RL efficiency optimization with a safe shadow → live rollout
- Stood up the on-prem stack (Docker, Keycloak, Grafana/Prometheus) and stress-tested Kafka toward ~100× scale (~1,100 controllers) to size hardware
- Designed the analytics data warehouse plus metadata-driven Airflow ETL; authored the platform HLD/LLD specs

Nesso — Farm-to-Fork Traceability AGRITECH

- Architected an **offline-first** platform (NestJS 10 + Next.js 15 + Expo monorepo, MongoDB/Redis) with SQLite-outbox sync and GPS polygon farm mapping
- Delivered public QR traceability, 12-language i18n, WCAG 2.2 AA, and 17-role RBAC across mobile, web, and QR portal (~99 screens)

Rangsons Aerospace (RAPL) — Enterprise Tools FULL-STACK · FE/UI-UX

- Built a 7-role **Customer RFQ→Quotation** platform with gated approvals (12 steps / 8 states / 4 gates), revision history, and ECharts reporting — React 18 + Node/Express + MongoDB
- Built a **Manufacturing Execution System**: shift/task tracking, dynamic sequence enforcement, lockout/orphan-task concurrency controls
- Owned front-end and UI/UX — a shared "Calm Enterprise" design system (light/dark) reused across RFQ, MES, and an IT Helpdesk console (kanban + SLA)
- Delivered an on-site AI-hardware suitability evaluation

IRIS — B2B Connected Fragrance-Diffuser Platform SYSTEM DESIGN · APP · BLE · UI/UX

- Delivered a **BLE fleet-ops platform** — React 19 admin + Expo/RN field app talking to hardware over Bluetooth, on Express/Socket.IO/MongoDB (Azure) — covering app development, multi-generation BLE device integration, and UI/UX
- Built the core loop: onboard → commission → consumption-ledger dry-out prediction → refill dispatch, with offline-capable field math
- Authored the **v1→v2 system redesign** — stateless Redis-backed Socket.IO, BullMQ workers, MQTT→time-series telemetry, Key Vault / OpenTelemetry / CI-CD, offline-first sync — pilot → enterprise-grade

Neural-Nexus (Nesso) — Hybrid RAG HEALTH

- Hybrid RAG combining **Neo4j + PostgreSQL** with LLMs — async FastAPI, parallel multi-source retrieval with custom merge/rank, SSE streaming, strict grounding, Redis/Valkey caching

Examic — CodeMint AI CODE-GEN

- NL/PRD → code → GitHub → Vercel/Azure deploy automation; OpenHands agent workflows; team formation & hiring

Freelance Software Developer

AUG 2022 – DEC 2022

- Delivered full-stack apps, dashboards, and legacy-system improvements for clients end-to-end, bridging directly into the Lingotran role

Veriteam Software Solutions

FEB 2021 – AUG 2022

Software Engineer — Full Stack

- Delivered 40+ production apps (CMS, CRM, e-commerce) — Java (Spring, Hibernate), JSP, JavaScript, Python
- Integrated Stripe, Razorpay, OAuth, JWT, RBAC; modernized legacy PHP/.NET; mentored 250+ students & interns

02 Open-Source Projects

Nexus Swarm ↗ repo

Stateful multi-agent engine on LangGraph (Manager-Worker) with Human-in-the-Loop breakpoints, PostgreSQL-checkpointed recovery; FastAPI + Next.js 14; Pydantic V2 typed state.

Grounded Voice Assistant ↗ [repo](#) ↗ [live demo](#) (🗣️ → Ask)

Retrieval-grounded assistant with voice I/O and page-level tool-calling. NDJSON token streaming from a Next.js route handler; browser-native STT/TTS via the Web Speech API (no vendor, no per-request cost); per-IP token bucket plus a daily budget guard; 3-model provider fallback so a retired model can't take it down; deterministic no-LLM retrieval path so it degrades instead of failing. Answers constrained to a fixed knowledge base to prevent fabricated claims.

DocMind AI ↗ [repo](#)

Hybrid RAG (dense + sparse) — FastAPI, ChromaDB, Llama3; multi-doc ingestion, semantic chunking, query-time reranking.

LiveNexus AI ↗ [repo](#)

Real-time speech — WebRTC → LiveKit SFU → Faster-Whisper with WebRTCvad gating & adaptive model selection; double-buffered live UI.

InferenceHub ↗ [repo](#)

Polyglot inference gateway — Node.js (I/O, Zod, auth) + Python (PyTorch/Scikit) over **gRPC + Protobuf** with shared .proto contracts.

VoiceSync AI ↗ [repo](#)

Privacy-first offline transcription — React → pre-signed MinIO → FastAPI webhook → Faster-Whisper; fully air-gapped.

03 Skills

LANGUAGES

JavaScript TypeScript Python SQL Java* Go*

BACKEND & APIS

Node (Express, NestJS) FastAPI Django REST GraphQL gRPC Microservices

VOICE AI

Whisper Faster-Whisper Deepgram ElevenLabs Kokoro TTS LiveKit WebRTCvad Web Speech API

DATABASES & VECTORS

PostgreSQL TimescaleDB MongoDB MySQL Redis Neo4j ClickHouse pgvector

FRONTEND

React Next.js 15 Tailwind shadcn/ui Framer Motion Zustand ECharts

AI / GENAI / NLP

OpenAI Claude Gemini Ollama / vLLM RAG (→ Agentic) LangGraph PydanticAI MCP SpaCy Tool-Calling Agents

DATA & STREAMING

Kafka Airflow WebSockets SSE Protobuf BACnet Offline-First Sync

DEVOPS · OBSERVABILITY · AUTH

Docker GitHub Actions AWS Azure Kubernetes Grafana Prometheus OpenTelemetry Keycloak

04 Education

MCA — NIE Institute of Technology, Mysore ·

2017–2020 · 8.2 CGPA

BCA — Vidya Vikas College, University of Mysore · 2014–2017 · 71%

05 Leadership

- Tech Lead & Organizer — 24-hour hackathon, MIT College, Mysore
- Trained interns across full-stack & AI/ML; led Examic-team developer hiring